

EDUCATION

University of Pennsylvania

Philadelphia, PA

M.S.Ed., Interdisciplinary Studies in Human Development, GPA: 3.94 / 4.00

August 2024

- Coursework: Cognitive Processes, Regression and Analysis of Variance (R), Brain-Computer Interface (Python)
- Scholarships: Penn GSE Merit Scholarship, Korean American Foundation Scholarship, Ok-Hanheum Scholarship

Handong Global University

Pohang, Korea

Bachelor of Science in Computer Engineering (GPA: 3.78/4.50)

August 2022

- Thesis: Analysis of the Relation Between Solar Wind and the Disturbance of the Geomagnetic Field
- Study Abroad: University of Mississippi, Multi-Disciplinary Studies, GPA 3.92 / 4.00
- Scholarship: 2018 Excellent Academic Record, 2019 Handong Alumni Association, 2020-21 Korea Rotary Foundation

RESEARCH EXPERIENCE

Boys Town National Research Hospital

Omaha, NE

Research Assistant/Cognitive Affective NeuroDevelopment in Youth Lab (PI: Giorgia Picci)

Sep 2024 –Jun 2025

- Led assistant for neuroimaging studies on puberty, social cognition, and mental health in autistic youth.
- Processed MRI and MEG data using FreeSurfer, Brainstorm, and BESA to extract cortical thickness, source localization, and time-frequency metric
- Coordinated participant recruitment, IRB documentation, and data integrity for longitudinal studies.

University of Pennsylvania

Philadelphia, PA

Research Assistant, Exploring Academic Resilience in Orchids (PI: Sean Plaskett)

Sep 2023–May 2024

- Conducted and transcribed 15 semi-structured interviews with high school students at risk of dropout to examine the relationship between stress, resilience, and academic performance.
- Utilized automated speech-to-text software (Otter.ai) and thematic coding to identify linguistic indicators of emotional distress and coping strategies.

Research Assistant/Possibility Mentoring (PI: Michael Nakkula)

Sep 2023–May 2024

- Designed and facilitated goal-oriented mentoring activities to support students at General George Meade School in identifying academic and career interests.
- Analyzed mentor-student interactions to identify relational, environmental, and emotional factors that enhance engagement and program effectiveness.

Seoul National University

Seoul, Korea

Research Assistant/Developmental Cognitive Neuroscience Laboratory (PI: Sangah Lee)

Oct 2022–Feb 2023

- Collected and analyzed children's episodic memory and multisensory learning data.
- Conducted assessments by using the Wechsler Intelligence Scale for Children (WISC) on 30 children.
- Analyzed eye-tracking data to model expectancy violation as a predictor of cognitive development outcomes.

University of Mississippi

Oxford, MS

Research Assistant/Cognition Underlying Behavior Laboratory (PI: Stephanie Miller)

Jan 2022–May 2022

- Led a preschoolers' cognition training experiment by applying mindfulness protocols at Lafayette School
- Recruited 50 children to participate by offering craft activities at the Mississippi Day and Oxford's farmers market

Handong Global University

Pohang, Korea

Research Assistant/Handong Artificial Intelligence Laboratory(PI: Charmgil Hong)

Mar 2019–Dec 2021

- Implemented artificial neural networks to predict the mechanical properties of stamping processes
- Created machine learning models to detect factors affecting pancreatic cancer incidence
- Designed a search engine about the Korean reunification through the analysis of web scraping
- Trained 40 lab members to gain the knowledge of developing prediction models by using Python

PUBLICATION

1. **Kwak Inhee**, P. Jihee, and H. Charmgil, “Analysis of the Relation Between Solar Wind and the Disturbance of the Geomagnetic Field”, *Korea Computer Congress(KCC), KCC 2020 Korea Computer Science Conference Papers*, July 2020, 1,581 - 1,583 (3 pages)

PRESENTATIONS

1. **Kwak Inhee**, “Understanding ADHD in Classroom: Insights from Educators” *the 1st Penn GSE Annual Student Showcase*, May 2024
2. A. Suhyun, **Kwak Inhee**, “Graph Neural Networks to project the Dehydrogenated Enthalpy of Liquid Organic Hydrogen Carriers(LOHC)” *HGU 2021 Capstone Festival Poster Presentation*, December 2021
3. **Kwak Inhee**, P. Jihee, and H. Charmgil, “Analysis of the Relation Between Solar Wind and the Disturbance of the Geomagnetic Field”, *Korea Computer Congress(KCC), KCC 2020 Korea Computer Science Conference Papers*, July 2020

TEACHING EXPERIENCE

Korean Language Instructor

Refugee Pnan & Jang-an Children Center

Seoul, Korea

Jan 2022–May 2022

- Taught Korean language through culturally relevant books to elementary students from multicultural backgrounds.
- Led weekly book discussions and reading comprehension sessions with ten middle and high school students.

Programming Instructor

Pohang High School, Daebang High School, Pohang City Disabled Welfare Center, Living Stone Academy

Korea & Nepal

2018–2020

- Instructed 90+ students in Python, Arduino, and MIT App Inventor.
- Supervised student projects such as website and game development to promote applied computational thinking.

Teaching Assistant, Physics and Open Source Software

Handong Global University

Pohang, Korea

Mar 2019–May 2020

- Provided weekly three-hour tutorial sessions, totaling over 40 hours per semester
- Evaluated and provided feedback on assignments and exams for 60 students to support academic improvement.

STEM Education Instructor, Korea Hydro Nuclear Power Global Program

Na Copchi, Beneshova, & Okrzyshki Schools

Třebíč, Czech Republic

Jun 2019

- Taught 60 elementary students fundamental principles of renewable energy by building solar-powered cars.
- Organized a Korean traditional cooking and cultural exchange workshop at Ibanchiche Deaf School Orphanage.

WORK EXPERIENCE

Applied Computing Foundation

Project Manager

Remote, San Francisco, CA

Jun 2021–Present

- Develop and implement curricula for Web Programming, Python, and Artificial Intelligence courses for K–12 and undergraduate students.
- Coordinate partnerships with local city centers to organize STEM-focused summer camps across the Bay Area.
- Authored and secured the 2025–26 *STEM Sprouts Grant* from the Glenn W. Bailey Foundation to expand STEM education in Central California.

People Space

Computer Engineering Intern

Irvine, CA

Jan 2020–Feb 2020

- Constructed a business prototype using 3D printing technology and integrated Python-based motor control systems.
- Gained hands-on experience in artificial intelligence model development, training real-world datasets for machine learning applications.

SKILL

- **Programming Language:** Python, R, Java, C, HTML, JavaScript, CSS, C++, SQL
- **Software:** Tableau, SPM, FreeSurfer, BESA, BrainStorm, Matlab